

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations
January 2023

19CS8X78 - DATA STRUCTURES AND ALGORITHMS

Duration: 3 Hours

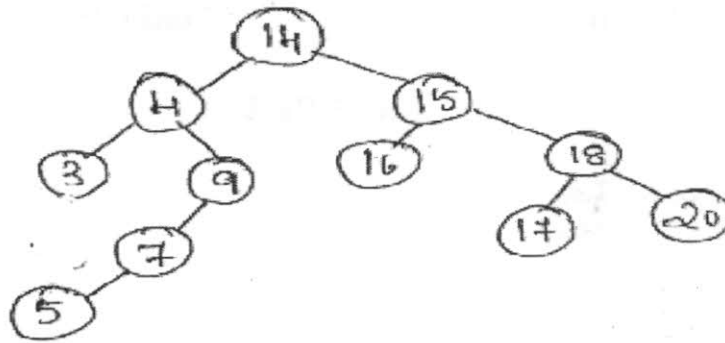
Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is Data structures? Explain with neat diagram classifications of Data structure.	7	L*1	1	1
	b) Define pointers. How to declare a pointer and accessing a variable through its pointer?	8	L1	1	1
	c) Differentiate Stack and Queue.	5	L2	2	1
2.	a) What is structure? How to declare a structure variable? How to access value from structure member?	8	L2	1	1
	b) Write a program for following operations of stack. i) push () ii) pop () iii) display () The program should check for stack overflow and underflow conditions.	9	L3	2	1
	c) What is Linear queue? What are the disadvantages of Linear queue?	3	L2	2	1
3.	a) Explain working of stack data structure with example.	6	L2	2	1
	b) Write a program for following operations on Linear queue. i) insert ii) remove iii) display The program should check for queue overflow and underflow conditions.	9	L3	2	1
	c) Write a C program to swap two numbers using call by address (pointers or reference) method.	5	L2	1	1
Unit – II					
4.	a) Explain various dynamic Memory allocation functions with syntax and examples.	9	L2	2	1
	b) Write C functions for the following operations on singly Linked List: i) Insertion at front end ii) Deletion from front end	6	L3	2	1
	c) Write note on the following terms with respect to tree. i) Leaf ii) Edge iii) Degree of a tree iv) Level of a tree v) Sub tree	5	L2	3	1
5.	a) Define Binary tree with an example. Write C recursive routine to traverse the given tree using inorder, preorder and postorder.	8	L1	3	1

P.T.O.

- b) Perform i) preorder ii) inorder iii) postorder tree traversal for the binary tree given below.



- c) What is doubly linked list? What are the advantages of double linked list over single linked list?

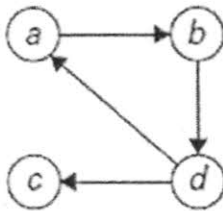
9	L3	3	1
3	L2	2	1

6. a) Explain adjacency matrix representation and linked list representation of graph with example.
 b) Discuss three asymptotic notations of algorithmic efficiency.
 c) Discuss the mathematical analysis of non-recursive algorithms.

6	L2	4	1
9	L2	4	1
5	L2	4	1

Unit – III

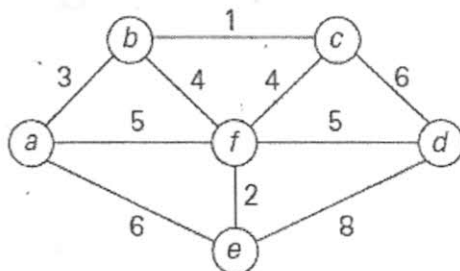
7. a) Explain BFS and DFS traversal technique with suitable examples.
 b) Apply Warshall's algorithm for below graph.



- c) Define dynamic programming.

9	L3	5	1
3	L1	5	1

8. a) Explain three types of decrease and conquer techniques.
 b) What is Prim's algorithm? Apply Prim's algorithm for below graph,



- c) Write a note on backtracking techniques.

8	L3	5	1
3	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
